THE ISOPERIMETRIC PROBLEM IN A PRIME BOX ON ONE PAGE

$2 \times 3 \times 5$ and the harmonic mean

A box's surface per unit volume is set by the harmonic mean of its sides, and its departure from a cube by how far that mean falls below the geometric one.

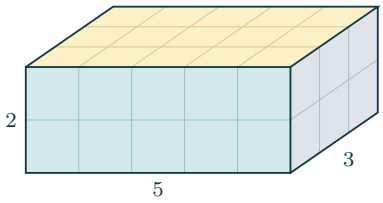
THE RULE

Surface to volume is six over the harmonic mean

A box with sides a, b, c has $V = abc$ and $S = 2(ab + bc + ca)$, so dividing turns the products into reciprocals. With the harmonic mean $H = 3/(\frac{1}{a} + \frac{1}{b} + \frac{1}{c})$ and the geometric mean $G = \sqrt[3]{abc}$,

$$\frac{S}{V} = 2\Big(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\Big) = \frac{6}{H}, \qquad \frac{S}{V^{2/3}} = 6\,\frac{G}{H} \geq 6$$

The first says every box has the surface-to-volume ratio of a cube of side H. The second, which does not change when the box is scaled, measures shape alone, and since $H \leq G$ with equality only when $a = b = c$, the cube uses the least surface for its volume among all boxes.



30 unit cubes behind 62 unit faces: $S/V = \frac{62}{30} = \frac{6}{H}$ with $H = \frac{90}{31}$.

THE SMALLEST PRIME BOX

$2 \times 3 \times 5$, the first box with three distinct prime sides

Its three means fall in the order the mean inequality demands,

$$H = \tfrac{90}{31} \approx 2.903 \;<\; G = \sqrt[3]{30} \approx 3.107 \;<\; A = \tfrac{10}{3} \approx 3.333,$$

and the shape measure is $G/H \approx 1.070$, so the box needs 7.0% more surface than a cube of the same volume, 62 against 57.93. Among the five boxes that 30 unit cubes can make, $S = 6V/H$ with V fixed means the largest harmonic mean wins, and the all-prime box wins outright.

In general $1 \times p \times qr$ loses to $p \times q \times r$ because their surfaces differ by $2p(q-1)(r-1) > 0$.

box	S	H
$1 \times 1 \times 30$	122	1.475
$1 \times 2 \times 15$	94	1.915
$1 \times 3 \times 10$	86	2.093
$1 \times 5 \times 6$	82	2.195
$2 \times 3 \times 5$	62	2.903

Every box of volume 30: surface falls as the harmonic mean rises.

OTHER PRIME BOXES

Close primes make nearly cubic boxes, but never a cube

Numbers pqr with three distinct prime factors are the **sphenic** numbers, and each builds exactly one box with prime sides. Because the sides always differ, G/H is always strictly above 1, and it approaches 1 only as the three primes crowd together relative to their size.

box	2×3×5	2×3×7	2×5×7	2×3×11	3×5×7	5×7×11	11×13×17	101×103×107
S/V	2.067	1.952	1.686	1.848	1.352	0.868	0.453	0.058
G/H	1.070	1.131	1.158	1.245	1.063	1.052	1.016	1.0003

Absolute size and shape separate cleanly: S/V falls steadily as boxes grow, while G/H ranks them by shape, so $3 \times 5 \times 7$ is rounder than $2 \times 3 \times 5$ even though both use three consecutive primes.

Why the harmonic mean

The harmonic mean averages quantities that add as reciprocals. Driving equal distances at 30 and 60 km/h gives an average speed of 40, the harmonic mean, because the times $1/30$ and $1/60$ per kilometre are what add. Each pair of faces contributes $2/a$, $2/b$ or $2/c$ of surface per unit volume, so the box's sides combine in the same way.

The isoperimetric problem

Among all solids of a given volume the sphere has least surface. The quotient $Q = 36\pi V^2/S^3$ equals 1 for a sphere and for a box becomes $Q = \frac{\pi}{6}\,(H/G)^3$, giving 0.524 for a cube, 0.427 for $2 \times 3 \times 5$ and 0.523 for $101 \times 103 \times 107$. In the plane, a $p \times q$ rectangle has $P/A = 4/H$ and the circle plays the sphere's part.

Conventions. For positive a, b, c the three means are $H = 3/(\frac{1}{a} + \frac{1}{b} + \frac{1}{c})$, $G = \sqrt[3]{abc}$ and $A = \frac{1}{3}(a + b + c)$, and they always satisfy $H \leq G \leq A$, with equality throughout only when $a = b = c$. Lengths are in units of one cube edge, so S counts unit faces and V counts unit cubes.

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